

1. Which of the following functions are performed by UDP?
- ☒ A. process-to-process communication
 - ☐ B. host-to-host communication
 - ☐ C. node-to-node communication
 - ☐ D. None of the choices are correct
2. The source port number on the UDP user datagram header defines _____.
- ☐ A. the sending computer
 - ☐ B. the receiving computer
 - ☒ C. the process running on the sending computer
 - ☐ D. None of the choices are correct
3. UDP packets have a fixed-size header of _____ bytes.
- ☐ A. 16
 - ☒ B. 8
 - ☐ C. 40
 - ☐ D. 32
4. TCP groups a number of bytes together into a packet called a _____.
- ☐ A. user datagram
 - ☒ B. segment
 - ☐ C. datagram
 - ☐ D. None of the choices are correct
5. TCP uses _____ to check the safe and sound arrival of data.
- ☒ A. an acknowledgment mechanism
 - ☐ B. out-of-band signaling
 - ☐ C. the services of another protocol
 - ☐ D. None of the choices are correct
6. In TCP, a SYN + ACK segment consumes _____ sequence numbers.
- ☐ A. no
 - ☐ B. three
 - ☒ C. two
 - ☐ D. one
7. A socket address is a combination of _____.
- ☐ A. A MAC address and a logical address
 - ☐ B. A MAC address and a port number
 - ☐ C. a user-specific address and a logical address
 - ☒ D. None of the choices are correct
8. _____ means accepting items from more than one source. _____ delivering items to more than one source.
- ☐ A. Demultiplexing; Multiplexing
 - ☒ B. Multiplexing; Demultiplexing
 - ☐ C. Encapsulation; Decapsulation
 - ☐ D. Pulling; Pushing
9. In TCP, if the ACK value is 200, then byte _____ has been received successfully.
- ☐ A. 199
 - ☐ B. 200
 - ☐ C. 201
 - ☐ D. None of the choices are correct



10. In TCP's _____ algorithm the size of the congestion window increases exponentially until it reaches a threshold.
A. congestion avoidance
B. congestion detection
C. slow start
D. None of the choices are correct
11. In the Go-Back- N protocol, the maximum send window size is _____ and the maximum receive window size is _____, where m is related to the number of bits in the sequence number.
A. 1, 1
B. 1, 2^m
C. comprehensive error control but limited flow control
D. All of the choices are correct
12. The network layer in the Internet provides _____.
A. comprehensive error and flow control.
B. limited error control, but no flow control.
C. comprehensive error control but limited flow control.
D. All of the choices are correct
13. In a datagram approach, the forwarding decision is based on the value of the _____ field in the packet header.
A. source address
B. destination address
C. label
D. None of the choices are correct
14. IP is a _____ protocol.
A. connection-oriented unreliable
B. connection-oriented reliable
C. connectionless unreliable
D. connectionless reliable
15. If the fragment offset has a value of 200 it means that _____.
A. the datagram has not been fragmented
B. the datagram is 200 bytes in size
C. the first byte of the datagram is byte 1600
D. None of the choices are correct
16. What is needed to determine the number of the last byte of a fragment?
A. offset number
B. total length
C. both offset number and the total length
D. None of the choices are correct
17. When a datagram is encapsulated in a frame, the total size of the datagram must be less than the _____.
A. MUT
B. MAT
C. MTU
D. None of the choices are correct
18. In subnetting, the number of addresses in each subnet must _____.
A. be a power of 2
B. be a multiple of 128
C. be divisible by 128
D. None of the choices are correct
19. What is the default prefix length for class C in CIDR notation?
A. 24
B. 8
C. 16
D. None of the choices are correct
20. The idea of address aggregation was designed to alleviate the increase in routing table entries when using _____ addressing.
A. classful
B. classless
C. classful or classless
D. None of the choices are correct
21. Which of the following is true about ICMP messages?
A. An ICMP error message may be generated for an ICMP error message.
B. An ICMP error message may be generated for a fragmented datagram.
C. An ICMP error message may be generated for a multicast datagram.
D. None of the choices are correct
22. Routing between autonomous systems is referred to as _____ routing.



☒ A. interdomain

B. intradomain

C. out-of-domain

D. None of the choices are correct

23. A one-to-many communication between one source and a specific group of hosts is classified as a _____ communication.

A. unicast

C. broadcast

☒ B. multicast

D. None of the choices are correct

24. Communication at the data-link layer is _____

A. end-to-end

C. process-to-process

☒ B. node-to-node

D. None of the choices are correct

25. Which error detection method consists of just one redundant bit per data unit?

A. Two-dimensional parity check

☒ C. Simple parity check

B. CRC

D. Checksum

26. In CRC, if the dataword is 111111 and the divisor 1010, what is the dividend at the sender?

A. 1111110000

C. 111111

☒ B. 111111000

D. 1111111010

27. If the sender is a host and wants to send a packet to another host on another network, the logical address that must be mapped to a physical address is _____

A. the destination IP address in the datagram header

C. the source IP address

☒ B. the IP address of the router found in the routing table

D. None of the choices are correct

28. In the _____ random-access method, stations do not sense the medium.

A. CSMA/CA

C. CSMA/CD

☒ B. ALOHA

D. Ethernet

29. The sender is a router that has received a datagram destined for a host in the same network. The logical address that must be mapped to a physical address is _____

C. the source IP address

D. None of the choices are correct

☒ A. the destination IP address in the datagram header

B. the IP address of the router found in the routing table

30. An ARP request is normally _____

☒ A. broadcast

C. unicast

B. multicast

D. None of the choices are correct

31. Given the Ethernet address 01011010 00010001 01010101 00011000 10101010 11001111 in binary, what is the address in hexadecimal notation?

A. 5A:88:AA:18:55:FC

C. 5A:18:5A:18:55:CF

☒ B. 5A:81:BA:81:AA:CF

☒ D. 5A:11:55:18:AA:CF

32. The _____ layer of Ethernet consists of the LLC sublayer and the MAC sublayer.

☒ A. data link

C. network

B. physical

D. None of the choices are correct.



33. Each station on an Ethernet network has a unique _____ address imprinted on its network interface card (NIC).
- A. 16-bit
 - B. 32-bit
 - C. 64-bit
 - ☒ D. None of the choices are correct.
34. Gigabit Ethernet has a data rate of _____ Mbps.
- A. 10
 - B. 100
 - ☒ C. 1000
 - D. 10,000
35. A VLAN as a local area network configured by _____.
- ☒ A. software
 - B. physical wiring
 - C. software or physical wiring
 - D. None of the choices are correct.

- 14 - How many hosts are on the network 172.16.41.0/28 B 16
- 16 - Communication channel shared by all the mechanism on the network Broadcast
- 13 - How many ~~Endless~~ ^{Collision} domain are created when you ... 12 port 1 2 Collision
- 1 - In ARP a packet is encapsulated directly into a ~~Data link~~ ^{Data link} Frames
- 2 - The proxy server reduces the load on the D original server
- 19 - the physical path between sender and receiver through which Transmission medium
- 18 - In computer network nodes are : the computer that originates the data C
- 17 - Which of the ip can be used as a loop back address : B 127.0.0.1
- 5 - which of the following proto is used to ip address and ethernet address ARP
- 4 - which of the following functionalities must be implemented by transport ~~layer~~ and network protocol : End to End connectivity A
- 22 - what are the methods to move data through a network of links and switches : packet and circuit switching D

24. A full-duplex communication channel permits information to travel
 a) At timed intervals
 b) One direction only
 c) Both ways, but not at once
 d) Both ways at once
25. Each IP packet must contain
 a) Only source address
 b) Only destination address
 c) Source and Destination address
 d) Source or Destination address
26. Which protocol does DHCP use at the Transport layer?
 a) IP
 b) TCP
 c) UDP
 d) ARP
27. There are the following statements that are given below, which of them are correct ARP?
 i. The ARP protocol is used to get destination MAC address from the destination IP address.
 ii. The ARP protocol is used to get the destination IP address from the destination MAC address.
 iii. The ARP stands for Address Resolution Protocol.
 iv. The ARP protocol works on the same network as well as connected networks
 a) i and iii
 b) ii and iii
 c) i, ii and iii
 d) i, iii and iv
28. Most packet switches use this principle
 a) Stop and wait
 b) Store and forward
 c) Store and wait
 d) Stop and forward
29. The required resources for communication between end systems are reserved for the duration of the session between end systems in _____ method.
 a) Packet switching
 b) Circuit switching
 c) Line switching
 d) Frequency switching
30. A local telephone network is an example of a _____ network.
 a) Packet switched
 b) Circuit switched
 c) Bit switched
 d) Line switched
31. What is main purpose of using subnetting?
 a) It divides a network into network classes
 b) It can speed up the network
 c) It divides one large network into several smaller ones
 d) None of the above
32. What does a VLAN do?
 a) Acts as the fastest port to all servers.
 b) Provides multiple collision domains on one switch port.
 c) Breaks up broadcast domains in a layer 2 switch internetwork.
 d) Provides multiple broadcast domains within a single collision domain.
33. What will be the subnet mask for the following IP address 100.101.110.100/11?
 a) 255.128.0.0
 b) 255.224.0.0
 c) 255.192.0.0
 d) 255.32.0.0
34. Which of the following the valid host range for the subnet on which the IP address 192.168.168.188 255.255.255.192 resides?
 a) 192.168.168.129- 190
 b) 192.168.168.128- 193
 c) 192.168.168.129- 192
 d) 192.168.168.129- 191
35. What is the broadcast address of prefix 172.18.16.0/22?
 a) 172.18.25.255
 b) 172.18.19.255
 c) 172.18.30.255
 d) 172.18.23.255
36. If you would like to give IP address to 30 hosts from your IP prefix which subnet mask you use most effectively?
 a) 255.255.255.128
 b) 255.255.255.240
 c) 255.255.255.224
 d) 255.255.255.252
37. There are the following statements that are given below, which of them are correct advantages of VLAN?
 i. Using VLAN, we can manage devices easily in the network.
 ii. Using VLAN, we can reduce the size of broadcast domains.
 iii. Using VLAN, we can add an additional layer of security.
 a) i, ii and iii
 b) i and ii
 c) i, ii and iii
 d) i, ii, iii



38. Which of the following types of VLAN connections are supported by Switch?
 a) Access Link b) Trunk Link c) A and B d) None of the above
39. There are the following statements that are given below, which of them are correct VLAN?
 i. VLAN is used to divide a physical network into multiple logical networks.
 ii. VLAN is used to break large broadcast domains into smaller broadcast domains.
 iii. VLAN stands for Virtual Local Area Network.
 iv. VLAN stands for Visual Local Area Network.
 a) i and iii b) ii and iii c) i, ii and iii d) i, ii and iv
40. What do you understand about the physical layout in network topologies?
 a) The physical layout that describes the physical placements of network devices.
 b) The physical layout describes the flow of data within the network.
 c) The physical layout that describes cables only used in the network.
 d) All the above

QUESTION2:

2.1 Suppose Host A sends two TCP segments back to back to Host B over a TCP connection. The first segment has sequence number 80; the second has sequence number 100. (2 MARKS)

- a. How much data is in the first segment?
 b. Suppose that the first segment is lost but the second segment arrives at B. In the response that Host B sends to Host A, what will be the acknowledgment number?

2.2 True or false? (3 MARKS)

- a. Host A is sending Host B a large file over a TCP connection. Assume that the data has been buffered in the sender. Host B will not send acknowledgments to Host A because Host B has received too much data. ()
 b. Suppose Host A sends one segment with sequence number 38 and 4 bytes of data. Host B receives the segment and sends an acknowledgment number 42. ()
 c. Suppose Host A is sending Host B a large file over a TCP connection. The sequence of bytes that A sends cannot exceed the size of the receiver's buffer. ()

What is the difference between a router and link-layer switch?

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Chapter: Transport Layer
Exercise: Review
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Question

Reduce

Suppose Host A sends two TCP segments back to back to Host B over a TCP connection. The first segment has sequence number 90; the second has sequence number 110.

- How much data is in the first segment?
- Suppose that the first segment is lost but the second segment arrives at B. In the acknowledgment that Host B sends to Host A, what will be the acknowledgment number?

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Answer



a)

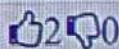
Consider sequence numbers, First segment=90

Second segment=110

Data in the first segment=110-90

=20

- b) Consider the first segment is lost but the second segment arrives at B. In the acknowledgment that Host B sends to Host A, then the acknowledgment number will be first segment of sequence number, that is 90.



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(Router)

connect various network
works on "Network layer"

used by (Lan) and (Man)

data is "packets"

Switch

connect // devices
// data link layer

used by Lan

// // Frames

139.103.2.1 ==> B

0 10 110 1110 1111

A B C D E

13.1.2.14 ==> A

A = 0 - 126 (127) ==> loopback

B = 128 - 191

C = 192 - 223

D = 224 - 239 multicast

E = 240 - 254

A ==> 10.0.0.0 - 10.255.255.255
B ==> 172.16.0.0- 172.31.255.255
C ==> 192.168.0.0 - 192.168.255.255